

# Evaluation report

Submission 1 · DCE

## SUMMARY

This report covers 1 submission comprising 16 parameter maps across DCE. No blocking errors or warnings were recorded. Reference maps were not available, so this report shows QC metrics only.

## METHODS

Each submission was checked for readable NifTI volumes and summarised by voxel-level statistics: the proportion of finite voxels, counts of NaN and infinite values, and the proportion of negative voxels.

No matching reference maps were available, so no agreement metrics were computed and only quality-control statistics are reported below.

All metrics carry the units of the parameter they describe and are never pooled across parameters or challenges.

Team and contact details were withheld from this report.

## SUBMISSION CONTENTS

| DATASET                            | TYPE | COUNT | DIMENSIONS | UNITS | STATUS |
|------------------------------------|------|-------|------------|-------|--------|
| Not available for this submission. |      |       |            |       |        |

Table 1. What the submission contains, grouped by dataset and type. Parameter maps, fitted signals and documents are counted separately; organiser reference data is not counted as submitted content.

## RESULTS

|                  |                 |
|------------------|-----------------|
| Submissions      | 1               |
| Maps             | 16              |
| Map types        | Ktrans          |
| Finite voxels    | 100%            |
| NaN / Inf        | 0 / 0           |
| Negative voxels  | Not available   |
| Map means        | Not available   |
| Reference status | ● Not available |
| RMSE             | Not available   |
| MAE              | Not available   |
| Bias             | Not available   |
| Spatial CoV      | Not available   |

Table 2. Aggregate quality-control statistics and reference agreement. Values are weighted across included maps; parameter types with different units are reported separately and never averaged together.

## RESULTS BY SUBMISSION

| SUBMISSION   | CHALLENGE | MAP TYPES | MAPS | FINITE VOXELS | NAN / INF | NEGATIVE VOXELS | MAP MEANS     | REFERENCE STATUS | RMSE          | MAE           | BIAS          |
|--------------|-----------|-----------|------|---------------|-----------|-----------------|---------------|------------------|---------------|---------------|---------------|
| Submission 1 | DCE       | Ktrans    | 16   | 100%          | 0 / 0     | Not available   | Not available | ● Not available  | Not available | Not available | Not available |

Table 3. Per-submission quality control and reference agreement. Measures that could not be computed are reported as Not available and are never converted to zero.

## ROI KTRANS STATISTICS

| DATASET   | PARTICIPANT | REPEAT | SITE | ROI        | MEDIAN | SD     | COV    | VOXELS | UNITS             | STATUS    |
|-----------|-------------|--------|------|------------|--------|--------|--------|--------|-------------------|-----------|
| clinical  | 1           | 1      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 1           | 1      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 1           | 2      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 1           | 2      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 2           | 1      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 2           | 1      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 2           | 2      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 2           | 2      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 3           | 1      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 3           | 1      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 3           | 2      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 3           | 2      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 4           | 1      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 4           | 1      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 4           | 2      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 4           | 2      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 5           | 1      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 5           | 1      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| clinical  | 5           | 2      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| clinical  | 5           | 2      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| synthetic | 1           | 1      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| synthetic | 1           | 1      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| synthetic | 1           | 1      | 2    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| synthetic | 1           | 1      | 2    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| synthetic | 1           | 1      | 3    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| synthetic | 1           | 1      | 3    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| synthetic | 1           | 2      | 1    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| synthetic | 1           | 2      | 1    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| synthetic | 1           | 2      | 2    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| synthetic | 1           | 2      | 2    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |
| synthetic | 1           | 2      | 3    | tumour     | 0.25   | 0.1118 | 44.72% | 4      | min <sup>-1</sup> | available |
| synthetic | 1           | 2      | 3    | brain mask | 0.45   | 0.2291 | 50.92% | 8      | min <sup>-1</sup> | available |

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Table 4. Within-ROI Ktrans statistics: 32 of 32 scan-ROI combinations available. ROI statistics were calculated from finite Ktrans voxels within each configured ROI. Standard deviation uses the population definition ( $ddof=0$ ). CoV is standard deviation divided by the absolute arithmetic mean and is unavailable when the mean is near zero. These conventions remain subject to final confirmation by OSIPI.

## ERRORS AND WARNINGS

| SEVERITY        | SUBMISSION | MESSAGE | AFFECTED FILE | RECOMMENDED ACTION |
|-----------------|------------|---------|---------------|--------------------|
| ● None recorded | —          | —       | —             | —                  |

Table 5. Errors and warnings raised during validation, with the action required before the submission can be shared.

## LIMITATIONS

- No additional limitations were reported for this export.
- Basic NIfTI QC checks readability and generic voxel statistics; it is not full BIDS validation.
- No official overall DCE score has been defined; no pass/fail scientific threshold is applied.
- Missing values are reported as Not available and are never converted to zero.